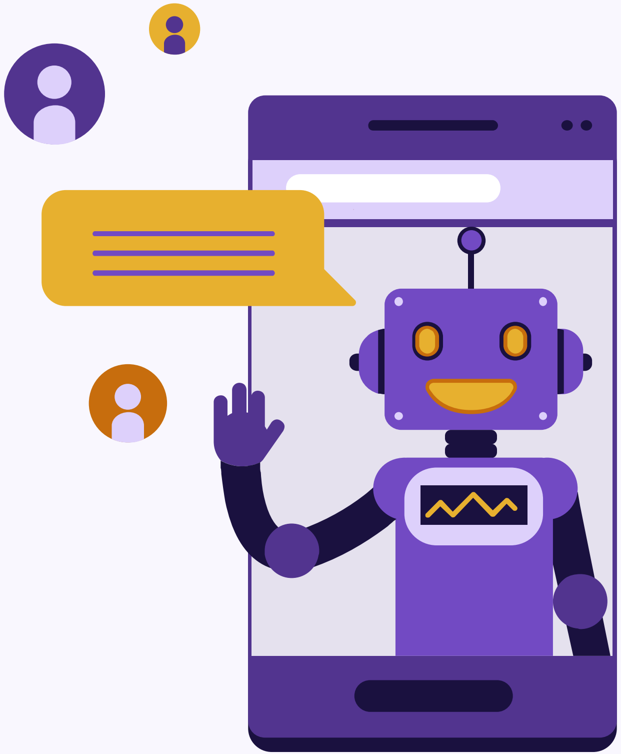
Characteristics of Artificial Intelligence in Scientific Writing

Jonathan, Leticia, Melle, and Wolter



Which is generated by an LLM?

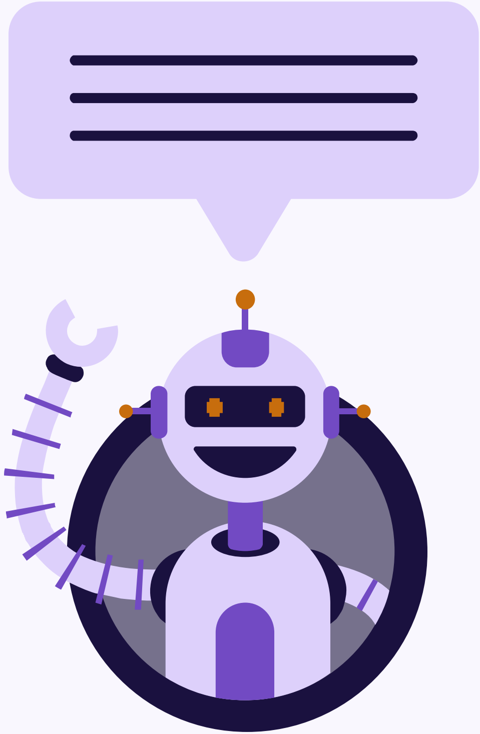
On the single-table complexity level with no join operations, SQL outperformed Polars for both aggregation tasks, obtaining significantly lower runtimes. These results were counterintuitive as Polars has the advantage of parallelization, which is often associated with faster runtimes. The slower runtimes observed for Polars are likely due to overhead related to the coordination of thread allocation and task scheduling required prior to running the query itself.

For straightforward, single-table aggregations that didn't require any joins, SQL surprisingly beat out Polars by delivering noticeably faster execution times. This outcome was unexpected because Polars utilizes parallel processing, a feature typically known for boosting performance. However, Polars likely suffered from initial lag caused by the system resources needed to manage threads and schedule tasks before actually running the code.

Introduction

- AI is now a co-author
- Writing made by LLMs goes unnoticed
  - **What characteristics of text are most indicative of it being AI-generated in scientific papers?**

“The rapid proliferation of sophisticated Large Language Models (LLMs) has blurred the lines between human and machine-generated text, posing significant challenges to academic integrity, content authenticity, and the combating of misinformation.” (Kingsley et al, 2025)



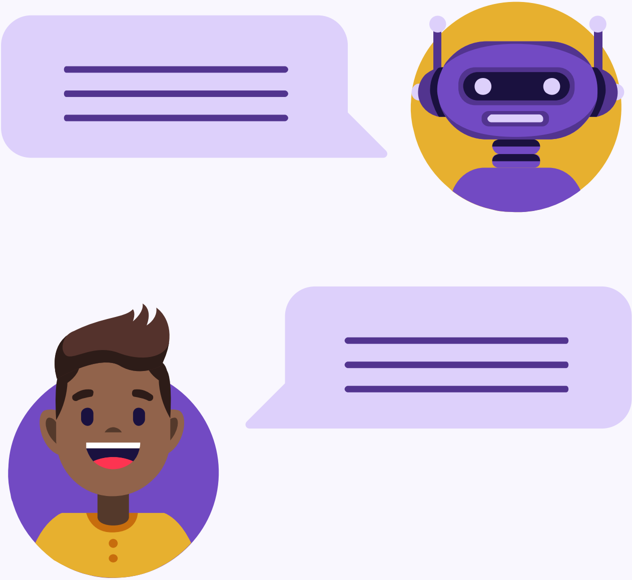
Dataset

Human Dataset

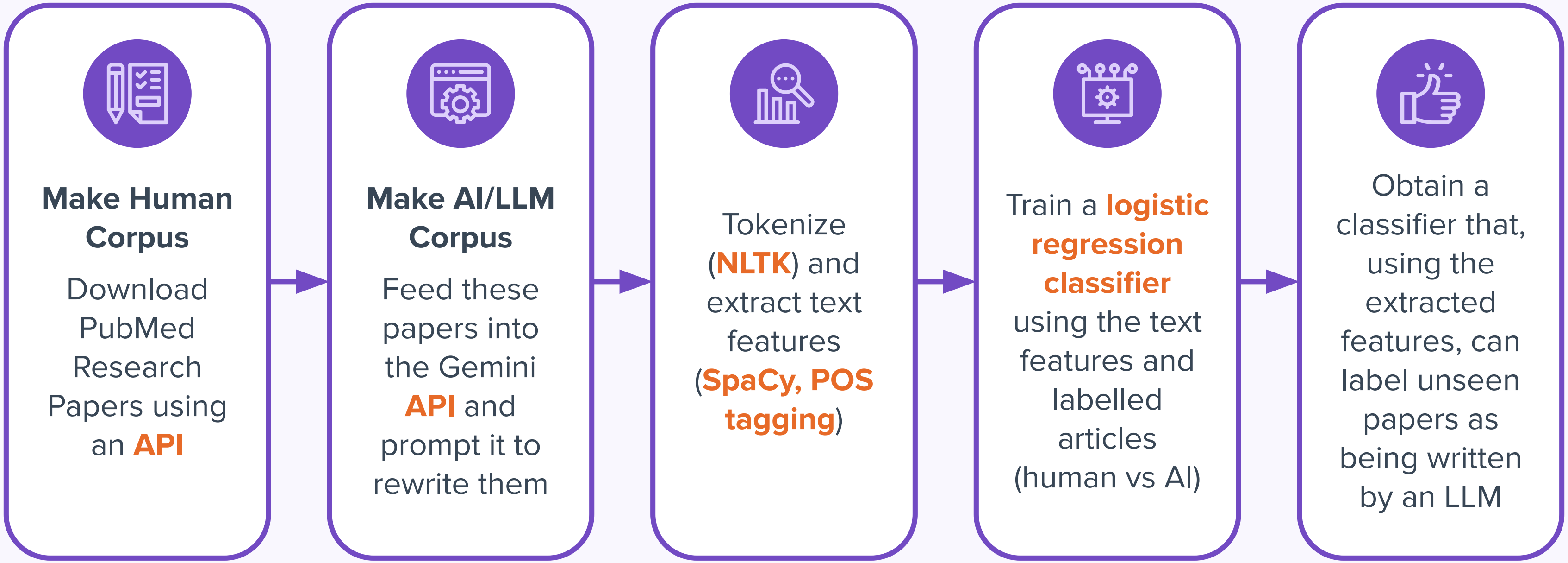
- 200 research articles from PubMed
    - Stored as .txt files
    - Only abstracts and paragraphs
    - Remove XML tags

AI Dataset

- Feed the articles into the Gemini API with a prompt to rewrite in
    - Paragraph by paragraph

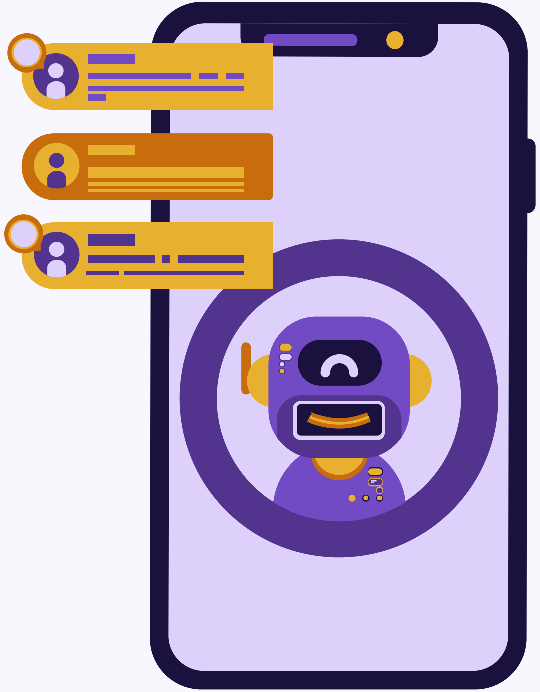


Methodology Pipeline



Methodology - Features

- Average sentence length
- Average word length
- Type to token ratio
- The variation of sentence lengths (burstiness)
- Average parse tree depth
- Percentage of words that are associated with AI writing according to the literature
- Percentage of words that we found to be used more in our AI (train) dataset
- **Function word counts:** Pronouns, adpositions, determiners, conjunctions, and auxiliary verbs.



Results

- **Macro-F1:** 0.854
- **Accuracy:** 0.854
- Decently balanced model

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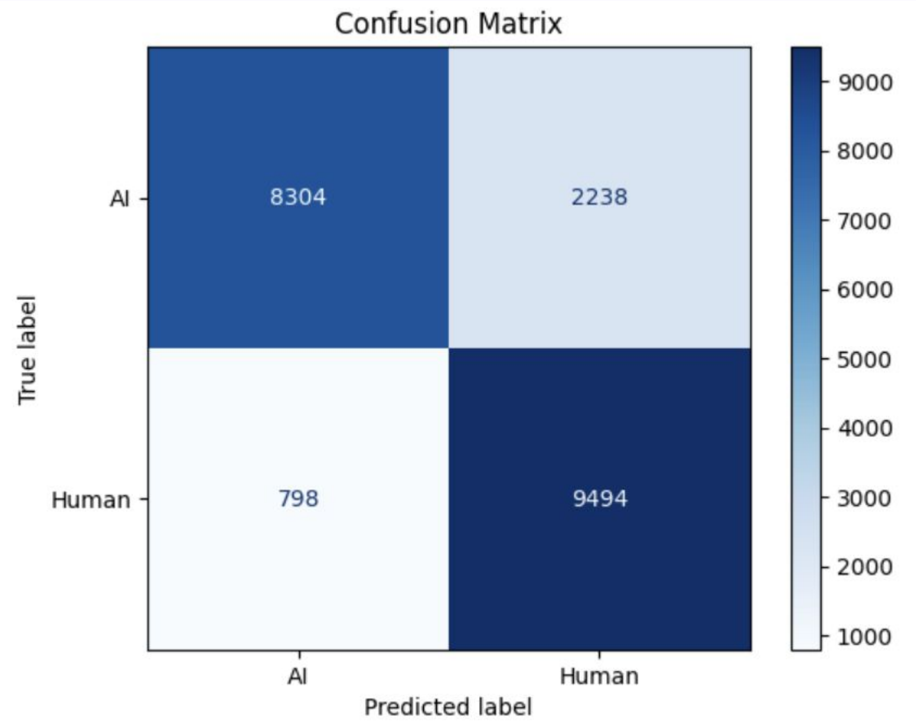
Feature Importances (Higher is more human-like):
    Burstiness                  +0.7324
    Auxiliary verb rate         +0.5168
    Av Parse Tree Depth         +0.4945
    Av Sentence Length          +0.3092
    Pronoun rate                +0.2343
    Conjunction rate            +0.0988
    Preposition rate            +0.0348
    Av Word Length              -0.0352
    Determiner rate             -0.3489
    Fixed AI Marker Rate        -0.3819
    Type/Token Ratio            -0.8501
    Learned AI Marker Rate      -2.8460

Intercept: [-0.22662397]
```



Feature Importance

- 3 most important features:
  - 1) Characteristic AI words (Learned AI Marker Rate)
  - 2) Type/Token Ratio
  - 3) Burstiness

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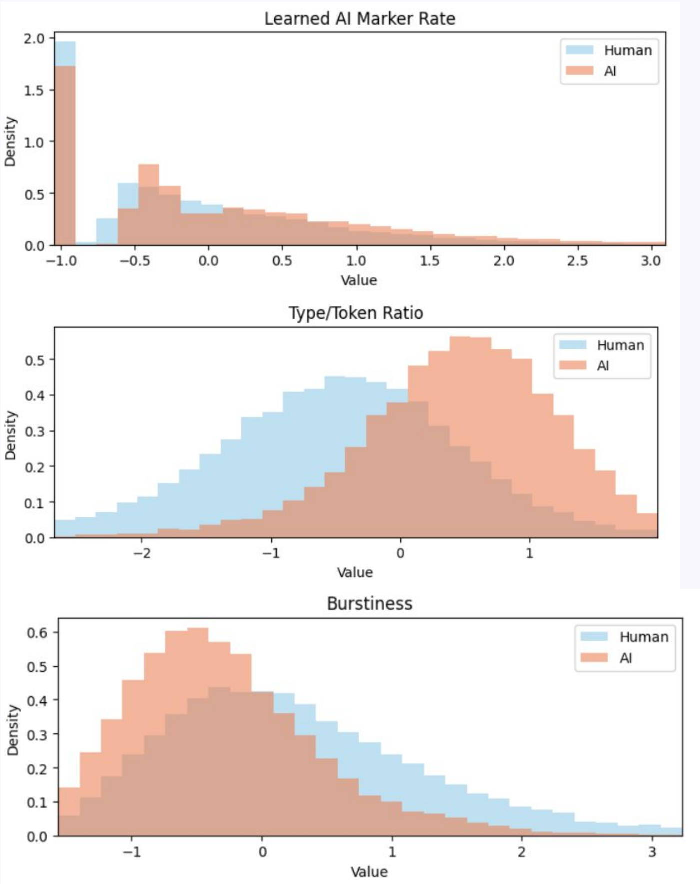


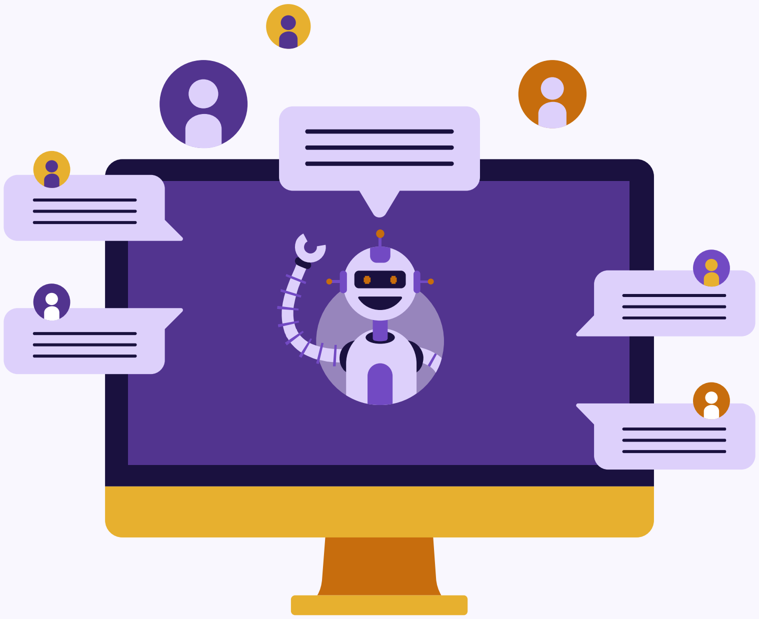
Limitations and Future Work

Limitations

- Relatively small dataset
- Solely PubMed articles
- Only Gemini was used to rewrite articles
    - Prompt used

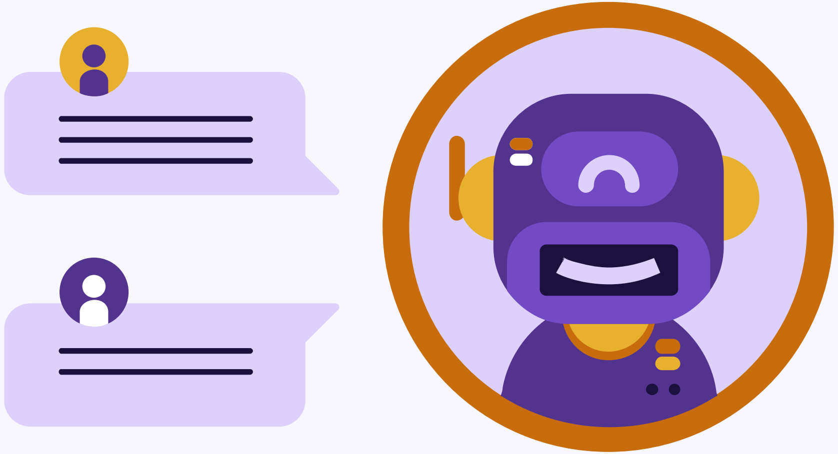
Future Work

- Naive Bayes
- Include other languages



Conclusion

- **What characteristics of text are most indicative of it being AI-generated in scientific papers?**
  - **Vocabulary**
  - Unique word use
  - Varying sentence length
- **Macro-F1: 0.854**
- **Decent and balanced prediction**



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